
Workshop 1: MSc thesis program

For a Master of Science (MSc) student in Software Engineering (SE) undertaking a practical thesis, whose project title format has the form "Develop a <technical> solution for a <business domain> problem".

1. Guidelines (c.f. Appendix B)

Section B of the paper's appendix provides a roadmap for researchers to move beyond a single study toward a robust **Research/Project Program**. The core philosophy is **Triangulation**: using different research strategies to overcome the inherent "three-horned dilemma" where no single study can maximize generalizability, precision, and realism simultaneously. The paper proposes two specific scenarios for designing a sequence of three studies.

2. Workshop Exercise

Step 1: Locate Your Topic

Decide if your research is **Knowledge-Seeking** (aiming to describe or explain a phenomenon) or **Solution-Seeking** (aiming to build a new tool or algorithm) .

Note: If you are building a tool (Solution-Seeking), you still need a Knowledge-Seeking study (like a Lab Experiment) to evaluate it!

Step 2: Map the ABC framework to your project

Create a table mapping each of the 8 archetypal research strategies of the ABC framework to your project.

Step 3: Design your 3-month MSc thesis plan

Draw a path between three strategies to form your program. Answer the following for your sequence:

1. Assume that you need 3 studies for your thesis project.
2. What research strategy do you plan to use for each study (see result of step 2)?

Step 4: Adapt the ABC-enabled MSc thesis template

Adapt the provided MSc thesis template to create a more refined template that incorporates the ABC-enabled thesis plan that you constructed in the previous step.

Step 4: Present and peer-review

Present your answers and critically review another team's answer.

- **The "Fairness" Check:** As a reviewer, identify the strategy your peer chose. Are you criticizing them for an **inherent** weakness? (e.g., "This lab experiment isn't realistic"). If yes, strike that comment—it is an unfair critique.
- **The "Strength" Check:** Did the student maximize the **potential strength** of their chosen quadrant? (e.g., "Is the measurement in this lab experiment actually precise?") .
- **The "Bridge" Check:** If there is a second study that serves as a bridge between two studies, does it actually help move to a different quadrant to provide a "holistic" view?.